



Contemporary · researchers including Roy Schwartz, Jesse Dodge, Noah Smith and Oren Etzioni.

"Efficiency must be a first-class metric."

THE WHY THAT DROVE THE WORK

They argue that the AI field's pursuit of accuracy at any computational cost — 'Red AI' — is unsustainable, and that efficiency should be a first-class research and business value.

INTELLECTUAL FOUNDATION

Their work documented the rising compute and carbon cost of state-of-the-art models and proposed efficiency as an explicit goal alongside accuracy.

FRAMEWORKS ANCHORED IN THIS COURSE

They anchor the Green AI vs. Red AI perspective informing Unit 3's cost, risk and sustainability Slibraries.

HOW THE IDEAS OPERATIONALISE

Their proposal operationalises as reporting and optimising the computational and environmental cost of models, not only their accuracy.

LIMITATIONS & CRITIQUE

Efficiency metrics are still inconsistently reported, and the trade-off between capability and cost is contested as models scale.

ENDURING IMPACT

'Green AI' helped put the energy and cost footprint of AI on the research and corporate agenda, influencing sustainable-AI practice.

ON THE SAME PANEL (UNIT 3)

Nassim Nicholas Taleb · Giant 11

Cathy O'Neil · Giant 12

Andrew Ng · Giant 13

J.B. Quinn · Giant 15

SOURCES (HARVARD REFERENCING STYLE)

Schwartz, R., Dodge, J., Smith, N.A. and Etzioni, O. (2020) 'Green AI', Communications of the ACM, 63(12).

Patterson, D. et al. (2021) 'Carbon emissions and large neural network training', arXiv:2104.10350.